

Inter-Annotator Agreement Analysis

This notebook analyzes the agreement between different annotators on the misinformation dataset. It uses the functions defined in `Interannotator_Analysis.py` to calculate and display agreement scores.

```
In [8]: # Add the 'src' directory to the system path to import our script
import sys
import os
# Get the current working directory of the notebook
notebook_dir = os.getcwd()
# Add the directory containing the script to the Python path
sys.path.append(notebook_dir)

# Import the functions from your script
from Interannotator_Analysis import get_dataframe, calculate_agreement, calc

# Import pandas for displaying DataFrames
import pandas as pd

# Configure pandas to display all columns
pd.set_option('display.max_columns', None)

print("Setup complete. Functions from Interannotator_Analysis.py are ready to use.")
```

Setup complete. Functions from Interannotator_Analysis.py are ready to use.

2. Load and Preprocess Data

Using `get_dataframe()` function from the Interannotator_Analysis script. This function handles finding all the `*annotations_cleaned.json` files, loading them, and concatenating them into a single, cleaned pandas DataFrame.

```
In [9]: # Load the data
df = get_dataframe()

print("DataFrame loaded successfully.")
print(f"Total annotations loaded: {len(df)}")
print(f"Number of unique annotators: {df['Annotator'].nunique()}")
print(f"Number of unique items: {df['ID'].nunique()}")

# Display the first few rows of the loaded data
df.head()
```

Found 5 cleaned annotation files.
All cleaned files have been loaded and combined.
DataFrame loaded successfully.
Total annotations loaded: 2033
Number of unique annotators: 5
Number of unique items: 1000

Out [9]:

	ID	Text	all_caps	exclamation_marks	hedging	adjectives	unk	Annotator
0	NaN	No Food, No FEMA: Hurricane Michael's Survivor...	[]	[]	[]	[]	[]	rachelle
1	1.0	Would-be looter in Hurricane Michael- ravaged F...	[]	[!]	[Would, would]	[]	[]	rachelle
2	2.0	His argument is correct. Hurricane Rita killed...	[]	[]	[]	[]	[]	rachelle
3	3.0	im praying for all of my friends down in the C...	[]	[]	[]	[]	[]	rachelle
4	4.0	To all my Texas streamers: PLEASE be safe if t...	[PLEASE]	[!]	[]	[]	[]	rachelle

3. Calculate Fleiss' Kappa (Overall Agreement)

Using the `calculate_agreement()` function from the `Interannotator_Analysis.py` file to compute Fleiss' Kappa. This metric is suitable for measuring agreement among multiple raters (more than two). Here, we are computing Fleiss's Kappa for four human annotators and 1 AI. The function automatically handles the calculation and prints a formatted table of the results.

```
In [10]: # Calculate and display Fleiss' Kappa
fleiss_results_df = calculate_agreement(df)
```

```

--- Fleiss' Kappa (Overall Agreement) ---
      Fleiss' Kappa      Agreement  Items
Feature
all_caps          0.499      Moderate    40
exclamation_marks  0.807  Almost Perfect    40
hedging           0.492      Moderate    40
adjectives         0.132       Slight    40
unk               0.440      Moderate    40

```

4. Calculate Cohen's Kappa (Pairwise Agreement)

Next, we compute Cohen's Kappa for every pair of annotators using the `calculate_pairwise_agreement()` function. This gives us a more detailed view of agreement between specific individuals.

```
In [11]: # Calculate and display pairwise Cohen's Kappa
pairwise_results_df = calculate_pairwise_agreement(df)

--- Cohen's Kappa (Pairwise Agreement) ---
              Gemini & Jennifer  Gemini & nicole  Gemini & rachel  Gemini & shiao-li
mini & shiao-li  Jennifer & nicole  Jennifer & rachel  Jennifer & shiao-li
nicole & rachel  nicole & shiao-li  rachel & shiao-li
Feature
all_caps          0.324          0.412          0.284
0.219             0.825          0.652          0.792
0.692             0.657          0.708
exclamation_marks  0.844          0.748          0.788
0.816             0.362          0.844          0.844
0.345             0.747          0.644
hedging           0.416          0.455          0.377
0.036             0.704          0.728          0.552
0.484             0.382          0.236
adjectives         0.101          0.127          0.105
0.018             0.487          0.381          0.487
0.131             0.657          0.418
unk               0.322          0.570          0.471
0.284             0.494          0.494          0.000
0.663             0.000          0.494
```

5. Visualize Agreement Scores

The bar chart below shows a comparison between the Fleiss' Kappa scores across different features.

Fleiss' Kappa Scores by Feature

```
In [12]: import altair as alt

# Check if the results DataFrame exists and is not empty
if 'fleiss_results_df' in locals() and not fleiss_results_df.empty:
    # Reset index to make 'Feature' a column for Altair
    fleiss_to_plot = fleiss_results_df.reset_index()

    # --- FIX: Rename column to remove apostrophe for Altair ---
    fleiss_to_plot = fleiss_to_plot.rename(columns={"Fleiss' Kappa": "Fleiss Kappa"})

    # Create the base bar chart
    bars = alt.Chart(fleiss_to_plot).mark_bar().encode(
        x=alt.X('Feature:N', title='Annotation Feature', sort=None),
        # Use the renamed column, but set a clean title for the axis
        y=alt.Y("Fleiss Kappa:Q", title="Fleiss' Kappa Score"),
```

```

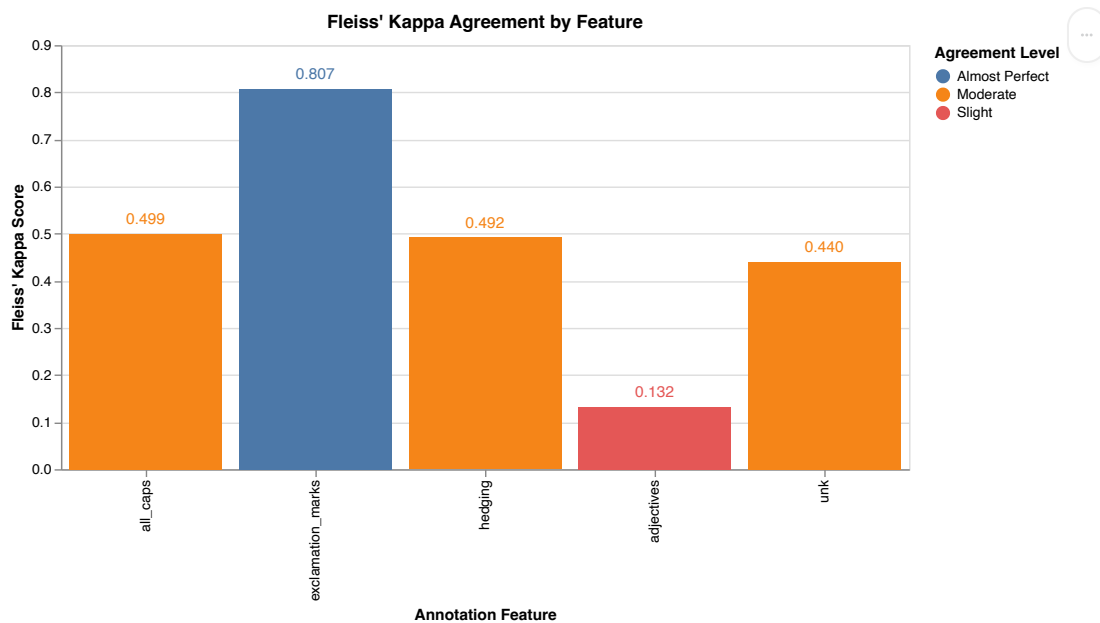
        color=alt.Color('Agreement:N', title='Agreement Level'),
        tooltip=[
            alt.Tooltip('Feature'),
            alt.Tooltip("Fleiss Kappa:Q", title="Fleiss' Kappa", format='.3f'),
            alt.Tooltip('Agreement')
        ]
    )

    # Create the text layer for labels on top of the bars
    text = bars.mark_text(
        align='center',
        baseline='bottom',
        dy=-5 # Nudge text up so it's not on the bar
    ).encode(
        # Use the renamed column for the text encoding
        text=alt.Text("Fleiss Kappa:Q", format='.3f')
    )

    # Layer the bars and text together
    chart = (bars + text).properties(
        title="Fleiss' Kappa Agreement by Feature",
        width=600
    )

    display(chart)
else:
    print("Fleiss' Kappa results are not available to plot.")

```



Pairwise Cohen's Kappa Scores

For the pairwise results, the following heatmap visualizes the agreement matrix between all pairs of annotators for each feature.

```

In [13]: # Check if the results DataFrame exists and is not empty
if 'pairwise_results_df' in locals() and not pairwise_results_df.empty:
    # --- FIX: Rename column to remove apostrophe for Altair ---

```

```

pairwise_renamed = pairwise_results_df.rename(columns={"Cohen's Kappa":

# Melt the DataFrame to make it suitable for Altair's encoding
df_melted = pairwise_renamed.reset_index().melt(
    id_vars='Feature',
    var_name='Annotator Pair',
    value_name="Cohen Kappa"
)

# Create the heatmap
heatmap = alt.Chart(df_melted).mark_rect().encode(
    x=alt.X('Annotator Pair:N', title='Annotator Pairs', sort=None),
    y=alt.Y('Feature:N', title='Feature', sort=None),
    color=alt.Color("Cohen Kappa:Q",
                    scale=alt.Scale(scheme='redyellowgreen'),
                    legend=alt.Legend(title="Kappa Score")
    ),

    tooltip=[
        alt.Tooltip('Feature'),
        alt.Tooltip('Annotator Pair'),
        alt.Tooltip("Cohen Kappa:Q", title="Cohen's Kappa", format='.3f')
    ]
)

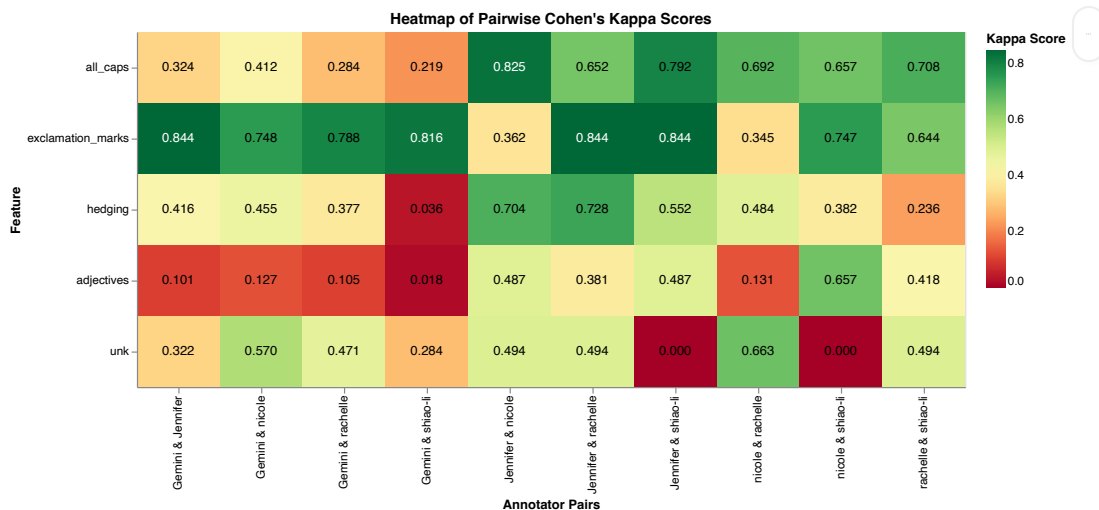
# Create the text layer for labels
text = heatmap.mark_text(baseline='middle').encode(
    text=alt.Text("Cohen Kappa:Q", format='.3f'),
    color=alt.condition(
        # Use white text for dark cells, black for light cells
        alt.datum["Cohen Kappa"] > 0.8,
        alt.value('white'),
        alt.value('black')
    )
)

# Layer the heatmap and text
chart = (heatmap + text).properties(
    title="Heatmap of Pairwise Cohen's Kappa Scores",
    width=700,
    height=300
)

display(chart)

else:
    print("Pairwise Cohen's Kappa results are not available to plot.")

```



6. Identify and Display Disagreements

To better understand the sources of disagreement, we are using a table that filters for only those items where annotators did not all provide the same annotation for a given feature.

The following code will:

1. Iterate through each unique item (`ID`) in the dataset.
2. For each item, check every feature to see if there is more than one unique value (i.e., a disagreement).
3. Collect all items that have at least one disagreement.
4. Display a table showing the annotations for these specific items, making it easy to see the differences in ratings.

```
In [14]: # Find items with any disagreement
disagreement_ids = []
features = ['all_caps', 'exclamation_marks', 'hedging', 'adjectives', 'unk']

# Group by item ID
grouped = df.groupby('ID')

for name, group in grouped:
    has_disagreement = False
    for feature in features:
        # Make a copy to avoid SettingWithCopyWarning
        feature_series = group[feature].copy()

        # Check if the column contains lists and convert them to sorted strings
        if feature_series.dropna().apply(lambda x: isinstance(x, list)).any():
            # The lambda function handles non-list values (like NaN) gracefully
            feature_series = feature_series.apply(lambda x: str(sorted(x)) if isinstance(x, list) else x)

        # Check if there's more than one unique value for the feature in this group
        if feature_series.nunique() > 1:
            has_disagreement = True
```

```

        break # Move to the next item as soon as one disagreement is found
    if has_disagreement:
        disagreement_ids.append(name)

# Filter the original DataFrame to show only the items with disagreements
disagreement_df = df[df['ID'].isin(disagreement_ids)].copy()

# Optional: Sort for clearer presentation
disagreement_df = disagreement_df.sort_values(by=['ID', 'Annotator'])

# To make disagreements more visible, we can pivot the table
# and display it. This puts annotators in columns.
if not disagreement_df.empty:
    print(f"Found {len(disagreement_ids)} items with at least one disagreement")

    # We can show a pivoted view for each feature to highlight disagreements
    for feature in features:
        # Create a temporary copy for this pivot to handle list-to-string conversion
        temp_df = disagreement_df.copy()

        # Convert list to string for pivoting if necessary
        if temp_df[feature].dropna().apply(lambda x: isinstance(x, list)).any():
            temp_df[feature] = temp_df[feature].apply(lambda x: str(sorted(x)))

        pivot_disagreements = temp_df.pivot_table(
            index=['ID', 'Text'],
            columns='Annotator',
            values=feature,
            aggfunc='first' # Use first since there's one value per annotator
        )
        # Filter to show only rows (items) that actually have a disagreement
        feature_disagreement_rows = pivot_disagreements.apply(lambda row: row[feature].isna())

        if feature_disagreement_rows.any():
            print(f"\n--- Disagreements for '{feature}' ---")
            display(pivot_disagreements[feature_disagreement_rows])
    else:
        print("No disagreements found across any items.")

```

Found 577 items with at least one disagreement.

--- Disagreements for 'all_caps' ---

		Annotator	Gemini	Jer
ID	Text			
6.0	WTF? » No food, no FEMA: Hurricane Michael's survivors are furious - The Daily Beast https://apple.news/APp4E5UMtQT2ULJVMPM0ovw ...		[]	['W
9.0	You know they hate you at work when the TV manager picks you to go out with a camera and operate the LIVE DRIVE car in a tornado.		['DRIVE', 'LIVE']	['D',
12.0	President Trump Gives Permission for US Troops to Stay at Trump Hotel in Washington DC https://t.co/PTHILtJqh3 via @gatewaypundit		['DC']	
14.0	"The effect of Hurricane Matthew on Haiti is catastrophic" - food + water crisis... http://ln.is/GSphX by #WSJ via @c0nvey		['WSJ']	
22.0	Drove thru NC last week & experienced my first tornado - went right in front of our vehicle, people on the hwy freaked out more than my kids		['NC']	
...	...		...	
988.0	I say this every hurricane: it's expensive AF to evacuate & not everyone has a reliable vehicle https://twitter.com/cnalive/status/901460592275275777 →†,Ä¶		['AF']	
989.0	78-Year-Old Woman, Scared to Drive in Blizzard, Found Dead in Car Outside NJ Burger King http://fb.me/4n2BfuH8G →†		['NJ']	
992.0	The potential for a border war that has been simmering at the Line of Actual Control (LAC) between India and China could reach a boiling point as hostilities increased on Monday night. https://t.co/5oPBrINKZB		['LAC']	
993.0	#CCP\n#Taiwan\nTaiwan has reportedly shot-down a Chinese Air Force Su-35 aircraft that supposedly violated its airspace\nh https://t.co/1cJyw2TZD2		['CCP']	
996.0	THR: DESTRUCTIVE TORNADO RIPPING THROUGH TUSCALOOSA, AL NOW! TAKE SHELTER NOW!!! #severe http://dlvr.it/PwRS5 (BN) #tcot		['AL', 'BN', 'DESTRUCTIVE', 'NOW', 'NOW', 'RIP...]	

159 rows x 5 columns

--- Disagreements for 'exclamation_marks' ---

ID	
1.0	Would-be looter in Hurricane Michael-ravaged Florida shot, killed after trying to steal law enforcement vehicle: report https://www.foxnews.com/us/would-be-looter-in-michael-ravaged-florida-shot-killed-after-trying-to-steal-law-enforcement-vehicle All looters need to be handled the same way.
4.0	To all my Texas streamers: PLEASE be safe if the hurricane comes to you! I, too, am a Texan. I don't want to lose...family...don't be a hero. Take shelter. pic.twitter.com/jDF8v8p1
23.0	*eats the medicine @mlp_twilight gives him* whats a solar flare anyways? don't worry about it.
48.0	Lately I been stressing make me wanna put a fuck nigga on a shelf.
49.0	Why are so many cars and buses stranded on the highway? Stay off the highway in the rain.
55.0	@jihettly @esd2000 good morning! Yes I'm in the middle of the blizzard. I have plenty of food and blankets. I'm happy! Lol. :)
58.0	Due to the poor air quality from the wildfires, our teams' Ride to Conquer Cancer was postponed yesterday. Big shout out to the Ski Cellar for still hosting our team for a BBQ yesterday. We could celebrate their training and fundraising successes. #therideAB pic.twitter.com/7v8p1jDF8
59.0	Re: Hurricane Matthew: All of the @ASUTennis Student-Athletes are on their way home. Thanks to @ASU_Housing for arranging a safe place for them.
81.0	Don't Panik! #KelbyTomlinson to the rescue! http://t.co/l3GizRUCy
87.0	Rapidly developing strong tornado over Clanton, AL crossing I-65. Take shelter in a basement or sturdy building.
90.0	Due to hurricane Matthew the Jacksonville workshop is being postponed due to the venue being used as a shelter. Stay safe.
103.0	Tornado warnings- welcome home Lauren. Take shelter in a basement or sturdy building.
109.0	Great Article @ReneMiamiHerald. Our answer is EVACUATE! Should you ride out or evacuate in a high-rise? http://www.miamiherald.com/news/business/real-estate/article168905202
115.0	Pep nt takin this hurricane seriously bt most of my stuff is packed and my dogs are packed so I can throw them in the car.
124.0	Just a reminder for anyone who wants to come to our Grand Re-Opening event after the October wildfires, it's tomorrow night from 4-7pm. Local food, local live music by The Awesome Hotcakes! See details below. :) pic.twitter.com/9v8p1jDF8
125.0	Oh dear!!!!!! Lizzard in Blizzard! Hahahahaha! I hope the sign didn't cause an accident. #2016snowstorm https://twitter.com/amnewyork/status/691009251125390080
134.0	Today is Corii Siren's birthday! Spoil her now: http://t.co/l3GizRUCy http://t.co/Hg8v8p1jDF8
136.0	Proud of my boss @TerronJernigan and 2 team members @IHXRex and @lhxBrownie for coming to the Panama City area to provide charging resources water/food and additior

ID	
	options for those affected by Hurricane Michael! # #ihxhttps://twitter.com/IHXRex/status/1050770676825
373.0	Shelter animals evacuate #Florida ahead of Hurricane #Dorian - Sep 2 @ 11:30 AM onsite!] https://oddcimes.com/static/2019/09/02/Shelter-animals-evacuate-Florida Hurricane-Do
398.0	I hope everyone in the path of hurricane dorian stays safe!! Don't hesitate to evacuate and if you are staying inside please remember to stock up on food and water. Send
413.0	So after these long tests, stuck in d building cuz of tornado and that also means no aunt! Just like yesterday! Fuckr
422.0	@crowdtappers @joinvroom OMG I remember the meltdown the day I did her hair like not ANNA.... OHHHH THE HORROR!!! LOL ;
432.0	Unlikely hurricane hero takes over chaotic Texas storm shelter: http://aol.it/2w @AOL\nZachary Dearir
442.0	@AbbigaleBailey Come to my house !! Or I'll ride that damn tornado to come get
517.0	This happen EVERY TIME it floods!! ,Äú@joeimel: BGFD dispatched to car stuck in behind McDonald,Äôs. The driver is still in
533.0	Hey #movietheatre mass murderer wanna-be we don't play that shit in
553.0	This wonderful lady is so right and so dead on in her description of Donald Trump s n will Love her! Democrats FAILED, they care about the inner city when its el
592.0	I wonder how @RoKhanna and @PramilaJayapal sleep at night when they have spoken out about what is being done to their Hindu sisters & brothers in Pakista girls are kidnapped, assaulted, lynchings of Hindu boys. Have they no co https://t.co/pY
609.0	Hello Emma, great to see that! It's spreading like wildfire! Hope the storage will be : and it will contribute to food and nutrition security and income. @ours
622.0	¡Desgraciados! No food, no FEMA: Hurricane Michael's survivors are furious - The https://apple.news/APp4E5UMtQT2ULJVM
692.0	Get comfy!! It's gonna be a rough ride!! (@ Frankenstorm Apocalypse - Hurricane Sar others) http://4sq.co
774.0	Ok, so now we're getting e-mails about how to seek shelter during a tornado? Dude, j
806.0	Where are Trump & Reps !?! Campaigning !!\nNo Food, No FEMA: Hurricane Michael' Are Furious https://thebea.st/2IV0u6C?source=twitter&via=desktop ... via @the
831.0	it's a whole flood in my
856.0	Hello Calgary!! Important news!! Please be advised!!! http://t.co/AR

ID	
864.0	Life Alarm donates supplies to shelter for Hurricane Matthew
873.0	@FunkyLilShack @mariaf30 I want a full on bitch slapping guns blazing cake throwi showdown!! Now THAT will be worth th
899.0	Get ur priorities str8 MIAMI!!! Yes issa Hurricane but what good is a full tank ofif dirty?ALWAYS stay CLEAN my friends! pic.twitter.com/Al
909.0	Hurricane evacuees, you can stay in a shelter with your pets in Austin. Austin Ani staff & volunteers are the https://twitter.com/austintexasgov/status/90219979189095
925.0	What a goooooooooa
928.0	@thoutaylorbrown I feel like accidents are just drawn to you but I'm happy you've sur them. Hope you
969.0	It looks like we re not the only ones who are 100% FED UP! with people protesting lawbreakers in our country Watch:Wow. Nasty Trump supporters in Iowa tear protestor s sign. RT @GoodSam10 @DREAMIowa pic.twitter.com/G5R6H1lLe8 (@TUSK81) September 12, 2015Here are the comments on the Twitter account where W
996.0	THR: DESTRUCTIVE TORNADO RIPPING THROUGH TUSCALOOSA, AL NOW! TAK NOW!!! #severe http://dlvr.it/PwRS5

--- Disagreements for 'hedging' ---

	Annotator	Gemini	Jennifer	nic
ID	Text			
1.0	Would-be looter in Hurricane Michael-ravaged Florida shot, killed after trying to steal law enforcement vehicle: report\n\n https://www.foxnews.com/us/would-be-looter-in-hurricane-michael-ravaged-florida-shot-killed-after-trying-to-steal-law-enforcement-vehicle-report ... All looters need to be handled the same way!	['Would', 'would']	[]	['Wol
4.0	To all my Texas streamers: PLEASE be safe if the hurricane comes to you! I,Ãd hate to lose...family...don,Ãt be a hero. Take shelter. pic.twitter.com/jDPPdMTHwN	['if']	[]	
15.0	i understand everyone has to make a buck, i would encourage you to think if food delivery during a blizzard is the best choice.	['if', 'think', 'would']	['think', 'understand', 'would']	['thi', 'understa', 'wol
16.0	Yeah, but it's also unthinkable to tell hurricane refugees to "have a great time," and to brag about the size of the crowd at shelter.	['about']	[]	
17.0	Birds in a blizzard. We put out extra sunflower seeds since their usual food sources just got,Ã¶ https://www.instagram.com/p/BBS2eYWgTbj/ ã†	['usual']	[]	
...	...	...	...	
987.0	i swea it feels like im about to explode ??	['about', 'feels', 'like']	NaN	†
990.0	@spinningbot Are you another Stand-user? If you are I will have to detonate you with my Killer Queen.	['If']	NaN	†
992.0	The potential for a border war that has been simmering at the Line of Actual Control (LAC) between India and China could reach a boiling point as hostilities increased on Monday night. https://t.co/5oPBrINKZB	['could']	NaN	†
994.0	I believe Donald's refusal to let Bahamian refugees from hurricane Dorian to leave the islands even temporarily to evacuate to the US would fall under the category of "attempted genocide." No power, no shelter, no medicine, little food or water—people *will* die.	['believe', 'would']	NaN	†
1000.0	NASA: RT NASAKennedy: With Hurricane Dorian forecast to bring hurricane conditions to the Kennedy Space Center in just a few hours, 120 members of the "Ride Out Team" reported to the Launch Control Center to	['possible']	NaN	†

		Annotator	Gemini	Jennifer	nic
ID	Text				
	monitor and mitigate possible damage to spacef... pic.twitter.com/cyqhcMPXRM				

186 rows x 5 columns

--- Disagreements for 'adjectives' ---

ID	Text	Annotator	Gemini
1.0	Would-be looter in Hurricane Michael-ravaged Florida shot, killed after trying to steal law enforcement vehicle: report\n\nhttps://www.foxnews.com/us/would-be-looter-in-hurricane-michael-ravaged-florida-shot-killed-after-trying-to-steal-law-enforcement-vehicle-report ...\\n... All looters need to be handled the same way!	['enforcement', 'enforcement', 'trying', 'tryi...	
2.0	His argument is correct. Hurricane Rita killed over 100 people on the road, drowned them in their own cars. pic.twitter.com/kulwIZW2EI	['argument']	
3.0	im praying for all of my friends down in the Caribbean who have no where else to go and are forced to ride through the hurricane. be strong https://twitter.com/ttrogdon/status/905205124699750401-†,Ä¶	['praying']	
5.0	RT @Stacy_Spencer: There is a Tornado warning. Service is canceled. Get to shelter and be safe. -´ praying for memphis.	['praying', 'warning']	
7.0	The aftermath of a hurricane is horrific. The heat/humidity is excruciating, no water/ice, no bathing, complete darkness, bugs, no warm food	['bathing', 'darkness', 'excruciating']	
...	...	...	
995.0	Kashmiris and Muslims protesting in Southall London UK after Namaz Juma agaisnt India's violation of Article 370 and called it "another Palestine alike in the making." https://t.co/JepImDzd2E	['making', 'protesting']	
996.0	THR: DESTRUCTIVE TORNADO RIPPING THROUGH TUSCALOOSA, AL NOW! TAKE SHELTER NOW!!! #severe http://dlvr.it/PwRS5 (BN) #tcot	['RIPPING']	
997.0	A two year old video is viral as Indians in Spain now are celebrating building of Ram Mandir in Ayodhya.\nA video is viral on social media in which a group of boys and girls in India attire using Dhols and other Indian musical instruments while walking on.. https://t.co/bWhO3OS4De https://t.co/dtd2CosbPe	['building', 'celebrating', 'using', 'walking']	
998.0	More Shelter Pets Coming To San Diego Due To Hurricane Harvey http://dlvr.it/PI0mGG-†	['Coming']	
1000.0	NASA: RT NASAKennedy: With Hurricane Dorian forecast to bring hurricane conditions to the Kennedy Space Center in just a few hours, 120 members of the "Ride Out Team" reported to the Launch Control Center to monitor and mitigate possible damage to spacef... pic.twitter.com/cyqhcMPXRM	['bring', 'possible']	

401 rows x 5 columns

--- Disagreements for 'unk' ---

		Annotator	Gemini	Jennifer
ID	Text			
6.0	WTF? » No food, no FEMA: Hurricane Michael's survivors are furious - The Daily Beast https://apple.news/APp4E5UMtQT2ULJVMPM0ovw ...		['WTF']	['WTF?']
26.0	Cars melt, power down as wildfire turns California town into ,Äòburning¬+hell,Äô https://www.siasat.com/news/cars-melt-power-down-wildfire-turns-california-town-burning-hell-1430954/¬+ ,Ä¶ pic.twitter.com/MEE6HbXWCs		['hell', 'hell']	[]
48.0	Lately I been stressing make me wanna put a fuck nigga on a stretcher!		['fuck']	NaN
68.0	We're ready for the hurricane and Stephen just left like tf are you doing????		[]	NaN
80.0	23 Killed, Cars Melt, Power Down as Wildfire Turns California Town Into ,ÄòBurning¬+Hell,Äô https://en.dhwanionline.com/23-killed-cars-melt-power-down-as-wildfire-turns-california-town-into-burning-hell/¬+ ,Ä¶		['Hell', 'hell']	NaN
83.0	Grocery employees in DC region are working their asses off to make sure folks have food/supplies ahead of blizzard. MVPs.\n\n#blizzard2016		[]	NaN
89.0	Yay now I get to ride my bike to school through ass crack of hurricane Matthew. I can hear wind howling from inside my house.		['ass']	NaN
156.0	Dear food lion fuckers: give me a hard time today and you will be under the blizzard		[]	NaN
374.0	,Äú@itsbritany_btch: Someone give me a ride home from school : (,Äù omg me too . Not trying to walk home during this tornado watch lmao		['omg']	NaN
375.0	AAYYEE real nigga! jp. RT: @CallMe_Eduardo Ay bruhh we really bout to ride out and find this baby ass whirlwind yall call a tornado		['ass']	NaN
393.0	It better not flood my car will be fucked		[]	NaN
396.0	If you wanna smoke cigs that's your own problem but when your breath smells like an old ash tray.. that's fucking disgusting		[]	NaN
409.0	@jvargyas55 @RandyGallina and I have to piss badly and flood has to shit, it's an all around horrible ride		['piss', 'shit']	NaN
413.0	So after these long tests, stuck in d building cuz of tornado and that also means no food from aunt! Just like yesterday! Fuckn tornado!!!		[]	NaN
422.0	@crowdtappers @joinvroom OMG I remember the meltdown the day I did her hair like ELSA and not ANNA.... OHHHH THE HORROR!!! LOL #tangletalk		['OMG']	NaN
478.0	i just drove with both my parents in the car lmao that was a panic attack waiting to happen		[]	NaN

		Annotator	Gemini	Jennifer
ID	Text			
494.0	We've been hearing about this blizzard for a week but tonight's snowfall came smooth TF outta nowhere. Cars just liked up on the road.		[]	NaN
543.0	Yesterday after getting back home after Hurricane Michael we were driving around looking for stations with gas and somewhere to get food then my hubby pulled out his bag of pumpkin reeses they may have been melted but omg it was just what I needed!		['omg']	NaN
548.0	The week I buy a new car, there's a fucking hurricane		[]	NaN
567.0	"Today, my dad decided to start preparing for Hurricane Sandy by buying \$300 worth of long-life and canned food. We live in Australia. FML"		['FML']	NaN
598.0	He's really concerned about the Americans in the path of Hurricane Dorian, so he invited them all to join him to play golf there to ride out the storm. I guessed they missed the message... Such a douche bag!		[]	NaN
616.0	After sacrificing thousands of lawn mowers my car has gas. Fuck this hurricane warning.		['Fuck']	NaN
650.0	Holy crap, it's all happening.\n Climate change, wildfires, hurricanes...\n Dictatorships, mass shootings...\n Fossil fuel supply, energy uncertainty...\n... all of which lead to food shortages, mass migration, refugees, civil unrest, civil war...\nGet ready!\n https://twitter.com/cnnbrk/status/1060956850634539008 →†,Ä¶		['crap']	NaN
662.0	I,Äöve been close to a wildfire not this large scale but at least this intense. It can be compared to crawling inside a high wattage microwave while trying to flee inside your cars. It felt like hell on earth.		['hell']	NaN
717.0	Twitter iTunes Royal Wedding Gasoline Cars Money Bieber Obama Politics Religion Tornado Insurance Medicine BULLSHIT #BotBait		[]	NaN
721.0	Aww lmao "you need to die" the only pplwho say shit like that on social media are 11 year old white boys with no love at home get a life\n\nWhat you so mad for ? you came for bcu while lookin like that\n\nBig ass nose could've been used as a shelter during hurricane Dorian		['ass', 'shit']	NaN
734.0	Wild Horses of North Carolina's Outer Banks Will Ride Out Hurricane Dorian 'Butts to the Wind'.\n\n https://news.yahoo.com/wild-horses-north-carolinas-outer-175048942.html # ...		[]	NaN
755.0	Due to all of the issues with Hurricane Michael, we will not be hosting the group ride this afternoon. It will be rescheduled ASAP We hope everyone is getting power back and staying safe!		['ASAP']	NaN

		Annotator	Gemini	Jennifer
ID	Text			
791.0	He's the HS athlete that includes his jersey number signing shit. Trump signs wall of shelter for hurricane victims http://thehill.com/blogs/blog-briefing-room/news/348996-trump-signs-wall-of-shelter-for-hurricane-victims →†,Ä¶		['shit']	NaN
867.0	New rule. No more fucking rallies or campaign events—by anyone—until these people get the help they need.\n\nNo Food, No FEMA: Hurricane Michael's Survivors Are Furious https://thebea.st/2IV0u6C?source=twitter&via=desktop ... via @thedailybeast		[]	NaN
873.0	@FunkyLiShack @mariaf30 I want a full on bitch slapping guns blazing cake throwing Charles showdown!! Now THAT will be worth the wait ????		['bitch']	NaN
912.0	Crazy video of Congressman Alan Grayson going off on a Politico reporter has been released. Grayson appears to be such an elitist and a pompous a@@ when addressing the reporter.		[]	NaN
941.0	Hurricane Harvey prepper update: Sh#t just hit the fan in South Texas... non-preppers hurting badly as food, ... - http://ronkollman.com/hurricane-harvey-prepper-update-sht-just-hit-the-fan-in-south-texas-non-preppers-hurting-badly-as-food-water-power-and-emergency-services-fail/ →†,Ä¶ pic.twitter.com/KoEOMZRpZ7		[]	NaN
971.0	@realDonaldTrump - Do your fvcking job! \nNo Food, No FEMA: Hurricane Michael's Survivors Are Furious https://www.thedailybeast.com/no-food-no-fema-hurricane-michaels-survivors-are-furious ...		[]	NaN
991.0	My car was just floating in this fucking flash flood. This is scary shit		['shit']	NaN